Increasing availability. Self-contained RF-certified module solutions that allow multiple protocols like TCP, UDP and IP on chip are flooding the market from various vendors. Built-in security features in these solutions help reduce certification times and allow the addition of communication to microcontroller based products. The ease of using these solutions is also increasing. Chauhan says, "The community approach allows you to take LoRa networks in your own hands. We have had successful implementations in about 100 cities with community participation."

Anand believes, "We need to move towards applied developments with open source hardware rather than just the hobby and fun part of it." Open source community has been helpful in supporting product development that could be made more useful with a focus towards product deployment and adoption, as it is finally critical for true success.

How secure are these systems

Security and analytics top the list of focus areas with IoT development. The IoT introduces a wide range of security risks and challenges to IoT devices, their platforms and operating systems, their communications and the systems to which these are connected. After all, experienced IoT security specialists are scarce, and security solutions are currently fragmented involving multiple vendors. As data volumes rise, newer analytic tools and algorithms are also needed, but as data volumes increase, traditional analytics are not enough to carry out the task.

At present, a very high number of IoT systems are open sourced and have open networks. This brings in the concern of security of these networks. These concerns are further fuelled by reports from Keeper Security, which highlight the state of passwords employed by users. Out of the one million passwords analysed by the group, a whopping 17 per cent went with the password 123456. Others in the list are no better. This brings in the question as to how secure the system can actually remain.

Improving the security. Security technologies required to protect IoT devices and platforms from information attacks and physical tampering must be improved significantly. Encrypting the communication and addressing innovative hacks and denial-of-sleep attacks that drain batteries must be looked at for the system to last longer. Security can also be made complicated by basic processors and simple operating systems used in edge devices that do not support complex security solutions.

Chauhan explains how they take care of security in their systems. "There are provisions for three levels of security in the systems. You need the network key that allows your device to connect with the network, the device key that allows you to access the device and the application key that allows you to access the particular data in the edge device." But if you set passwords like 123456, the system is going to have a tough time keeping intruders out.

Some ways to improve security can be implemented throughout the development cycle, starting with a threat model to consider the dangers and designing the system accordingly. Threat modelling helps the design team to consider vulnerable factors while designing the system, instead of post deployment. Security can then be implemented throughout the device lifecycle to keep those hackers on their toes.

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TABLE II
RECENTLY-LAUNCHED EDGE MODULES TARGETING THE IoT

Name	Supporting technology
RF-Lora from RF solutions	LoRa
CMWX1ZZABZ-078 from Murata	LoRa
MultiConnect xDot from Multi-Tech Systems	LoRa
ATA8520-EK from Microchip	Sigfox
Toby R2 series from U-blox	LTE
Monarch from Sequans Communications	LTE-M
Cinterion EMS31 from Gemalto	LTE-M
SARA-N2 from U-blox	NB-IoT
EYSHSNZWZ from TAIYO YUDEN	Bluetooth

What to expect in the future

Multiple ecosystems are expected to be the norm in the future, so interaction between different ecosystems is going to be another thing to look at. Areas such as smart homes, smart cities and healthcare call for robust systems with more reliability. Developers must regularly update the systems as more standards evolve in the future.

All things considered, multiple protocols could eventually help you out. There is Amazon Echo Dot competing with Google Home. Simply saying the words could get your prescription ordered or your air quality monitored. We have somehow managed to add connectivity in everything from televisions to tyres.

Considering a mix of image analytics and reactions on these by mechanical systems could be the next big thing. Adding image analytics to mechanical systems and tweaking some algorithms could lead to a much easier lifestyle. Which brings in the question, "Would that level of comfort be good for us and not turn us into lazy people sitting and typing everything away?" EFY